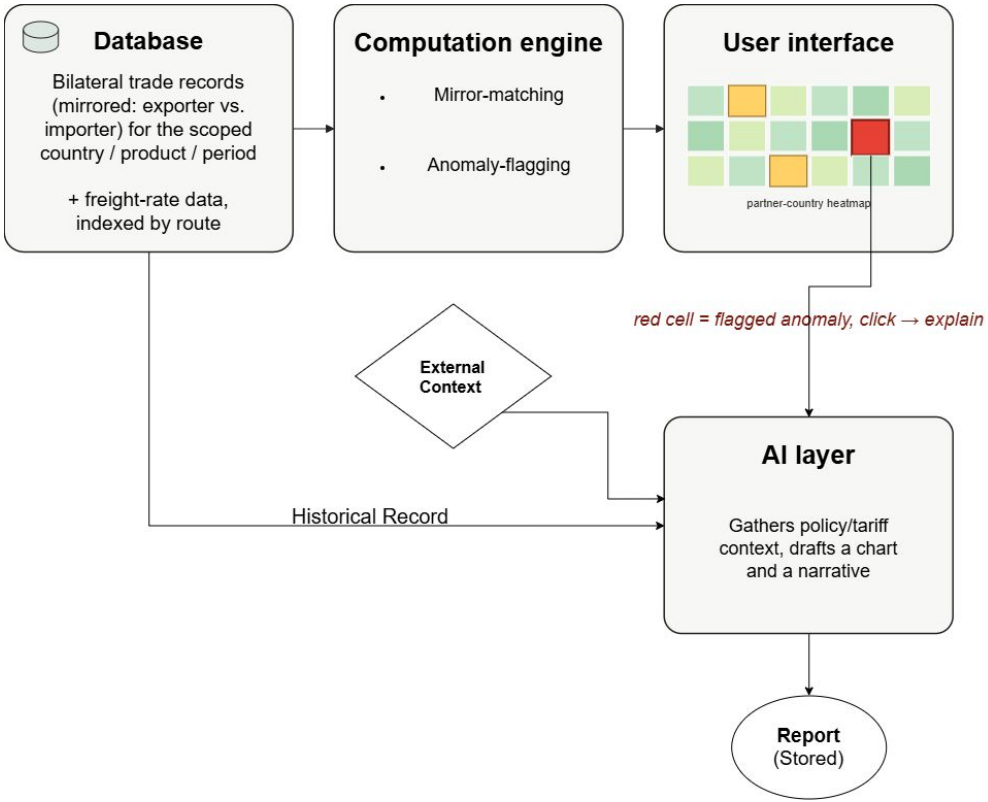
Project 3: Cloud-Native Trade Anomaly Detection & Reporting Platform

T-CloudSys: Cloud Services & Systems · 2026 – 2027

Project 3: Cloud-Native Trade Anomaly Detection & Reporting Platform

- Client : https://www.dnext.io/
- Key contact: Nabil Abdennadher, Omar Bouaziz, Sami Jaballah (DNEXT)
- Ojectives:
  - Design and implement a cloud data pipeline that ingests customs and freight data (transportation costs) into a managed data warehouse.
  - Deploy a backend service that runs the provided mirror-matching and anomaly-flagging computation at scale and exposes it through a documented API.
  - Deploy a dashboard that visualizes flagged anomalies.
  - Integrate a generative AI layer that turns a single flagged anomaly into a structured, narrative report.

Project 3: Cloud-Native Trade Anomaly Detection & Reporting Platform



Data: three tiers of trade sources in DNEXT

In reality, DNEXT draws trade data from three tiers, from national Authorities to Third party Tiers

TIER 1

National authority

- Each country's customs or statistics office
- Scope: one country
- Finest detail: national 8 to 10-digit codes

TIER 2

Eurostat

- EU statistical office (Comext database)
- Scope: EU member states
- Harmonised CN 8-digit codes

TIER 3

UN Comtrade

- UN global trade database
- Scope: worldwide
- HS 6-digit codes, comparable across countries



Coverage widens and product detail narrows from tier 1 to tier 3

Data: UN Comtrade

- **Free, official UN database** of international trade, by reporter, partner, product (HS code) and period
- **Your job:** download the data you need and load it into a database of your choice

Workflow

Select on Comtrade → **Download Excel** → **Load into your DB**

Key columns

Reporter | Partner | Flow | HS code | Period | Trade value | Net weight

Sample of a Comtrade download: HS 1001 (wheat), 2023

Period	Reporter	Flow	Partner	HS	Trade value (US$)	Net weight (kg)
2023	France	Export	Egypt	1001	412,500,000	1,650,000,000
2023	Egypt	Import	France	1001	470,250,000	1,700,000,000
2023	France	Export	Morocco	1001	318,000,000	1,200,000,000
2023	Morocco	Import	France	1001	336,000,000	1,150,000,000
2023	France	Export	Algeria	1001	540,000,000	2,000,000,000
2023	Algeria	Import	France	1001	810,000,000	2,100,000,000

Each pair of rows is the same flow reported twice: by the exporter (FOB) and by the importer (CIF).

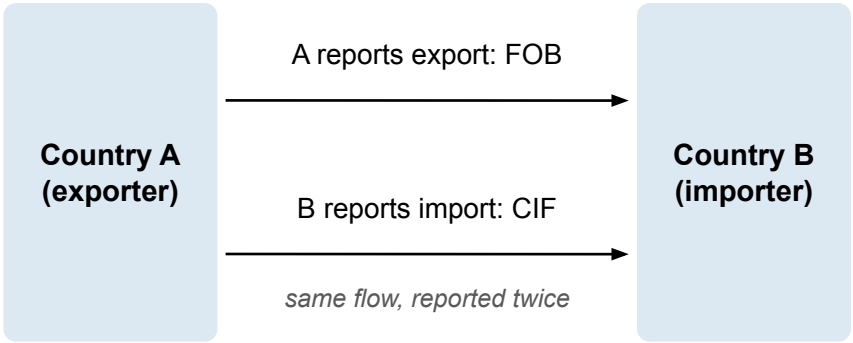
Illustrative data, not real figures

Key concepts: unit price, FOB, CIF

- **Comtrade gives totals, not prices:** you compute the unit price yourselves
- **FOB (Free On Board):** unit price declared at export
- **CIF (Cost, Insurance, Freight):** unit price declared at import
- **CIF should normally be slightly higher than FOB**
- **Mirror matching:** A's exports to B (FOB) are compared with B's imports from A (CIF)

Unit price = Trade value ÷ Quantity (e.g. USD/kg)

CIF = FOB + Freight + Insurance



The anomaly rule

$$\text{Gap} = (\text{CIF} - \text{FOB}) / \text{FOB}$$

Case 1: FOB > CIF
Anomaly. The import price cannot be lower than the export price.

Case 2: Gap > 35%
Anomaly. Freight and insurance cannot explain such a large difference.

Otherwise
Consistent flow. Nothing to flag.

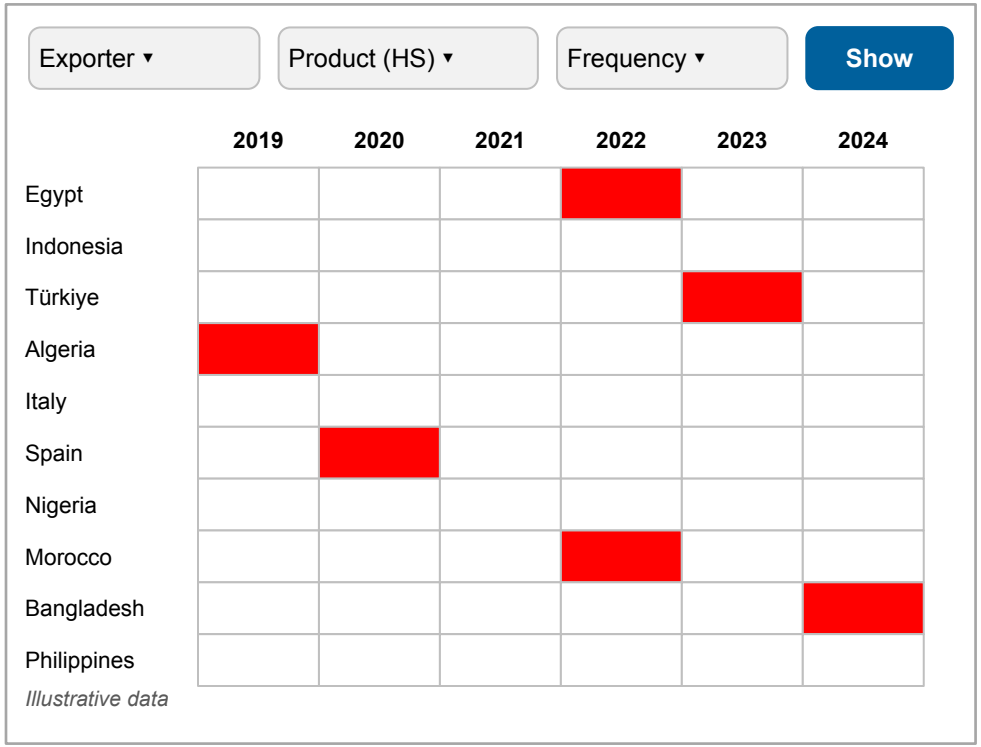
Example with FOB = 400 USD/t

CIF	Gap	Result
440	+10%	OK
600	+50%	**Flagged (> 35%)**
380	−5%	**Flagged (FOB > CIF)**

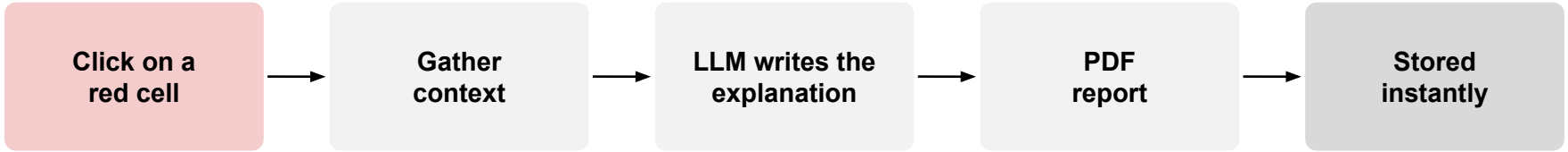
Threshold fixed at 35%

User interface

- **Two sections:** configuration panel + heatmap
- **Configuration:** exporter, product (HS code), frequency
- **Rows:** top 10 partner countries
- **Columns:** periods (yearly, quarterly or monthly)
- **Cells:** white = consistent, red = flagged
- **Red cell is clickable** and triggers the AI explanation



AI layer and reporting



Anomalies often have a reason

- Rerouting via third countries
- War and conflicts
- Tariffs and sanctions
- Data-entry errors

What we expect

- **A generative AI workflow** triggered by the click
- **Context:** historical data from your database + external sources
- **Asynchronous, multi-step job,** not one blocking call
- **Report stored instantly** and retrievable later

Your job: choose, build, justify

You choose the stack

Database
relational, data warehouse, …?

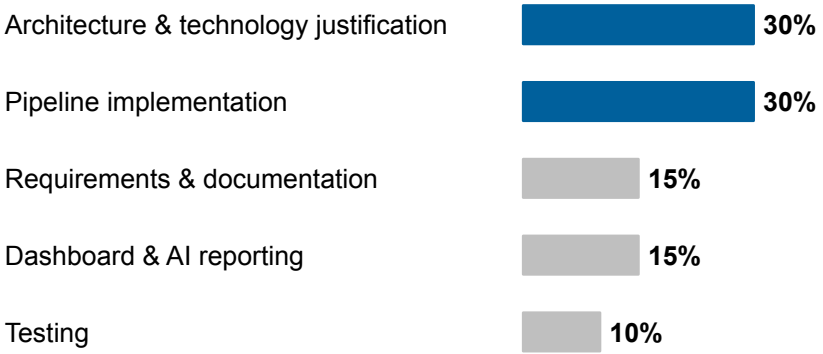
Compute
serverless or containers?

UI hosting
deployed independently

LLM & storage
which model, where to keep reports?

- **Every choice must be justified** (30% of the grade)

Grading



Key dates

6 Oct, 8:45	Project selection deadline & start
3 Nov, 8:45	Final submission

Next step (next week : 13th October 2026)

- Propose the global architecture of the platform
  - Which component running on which infrastructure ?
  - What cloud services to use ?
- Keep in mind that DNEXT is working with AWS Amazon …